

Horse

id
herdId
Speed

shadowLinkPtr
keyForSlaves

masterLinkPtr
keyToMaster

sumOfSlaves
flagFoundIter

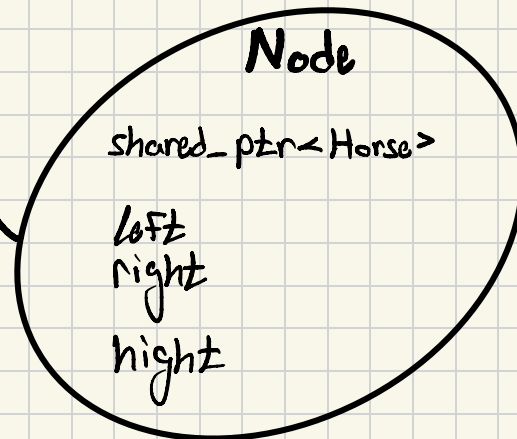
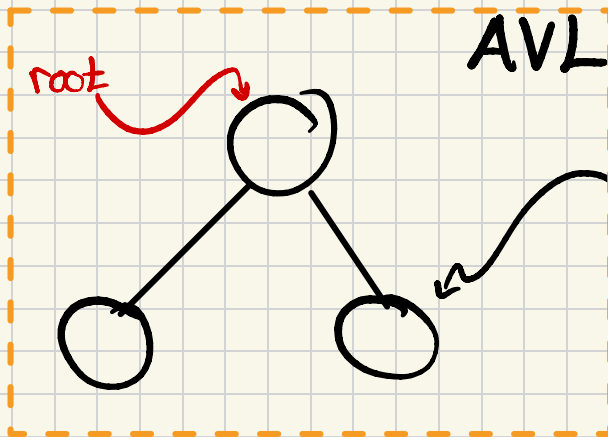
HorseLink

id
herdId

originHorsePtr
publicKey

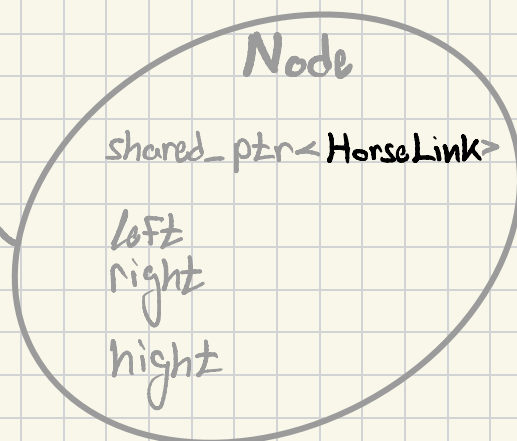
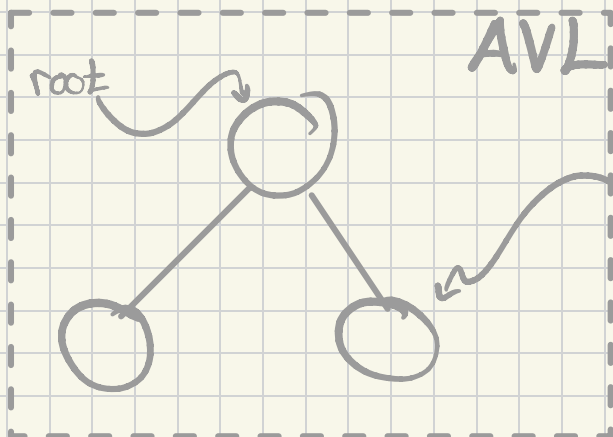
this is how
we secure our connections.

AVLTree < shared_ptr<Horse> , compareFunctor > allHorses



*sorting by
horse ID
done using
compareFunctor

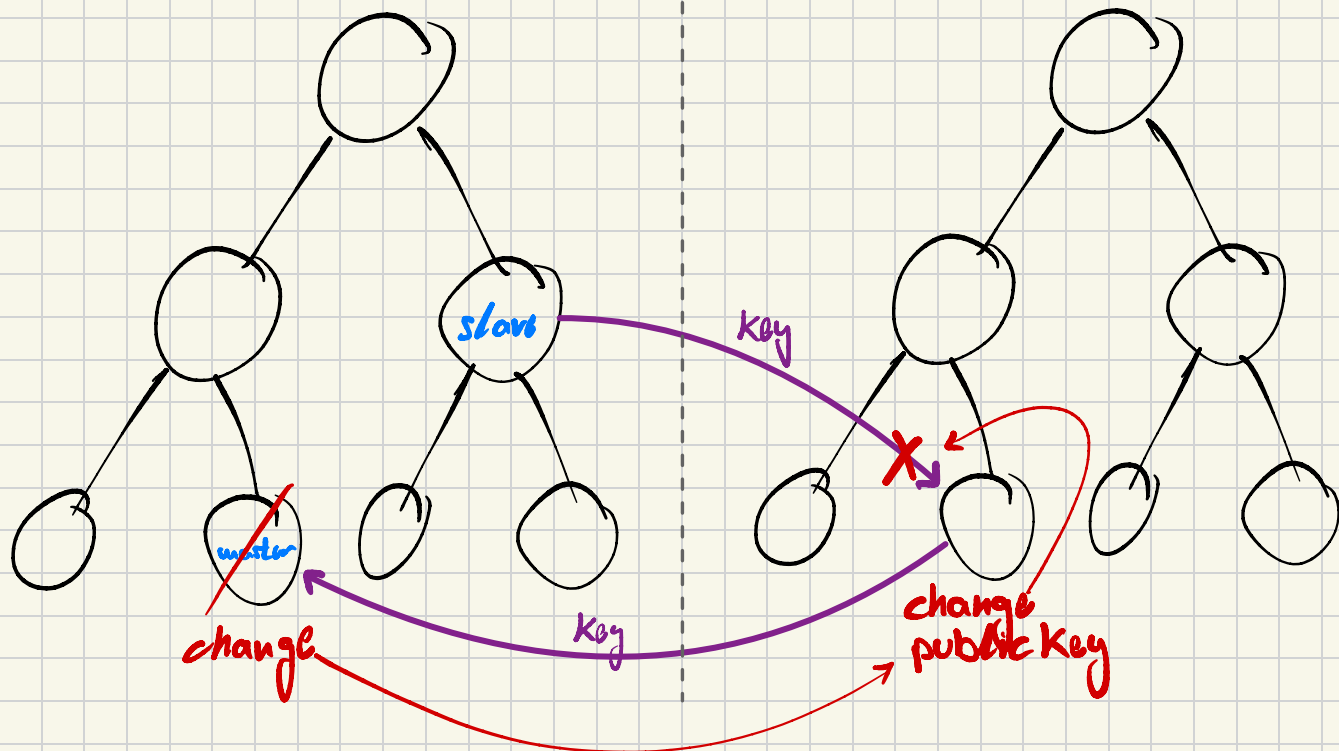
AVLTree < shared_ptr<Horse> , compareFunctor > allLinks



*sorting by
horse ID
done using
compareFunctor

all Horses

all Links

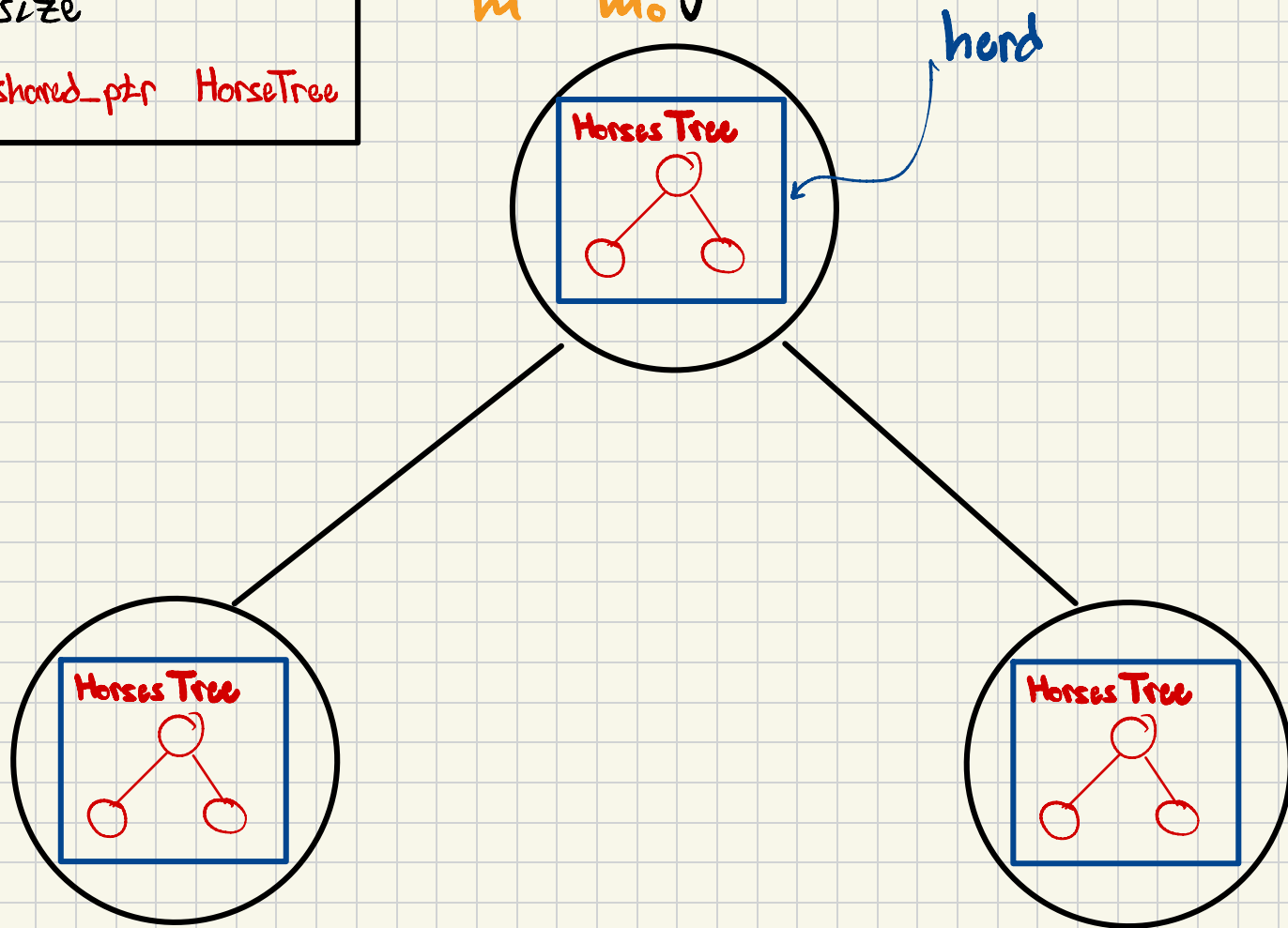


Herd

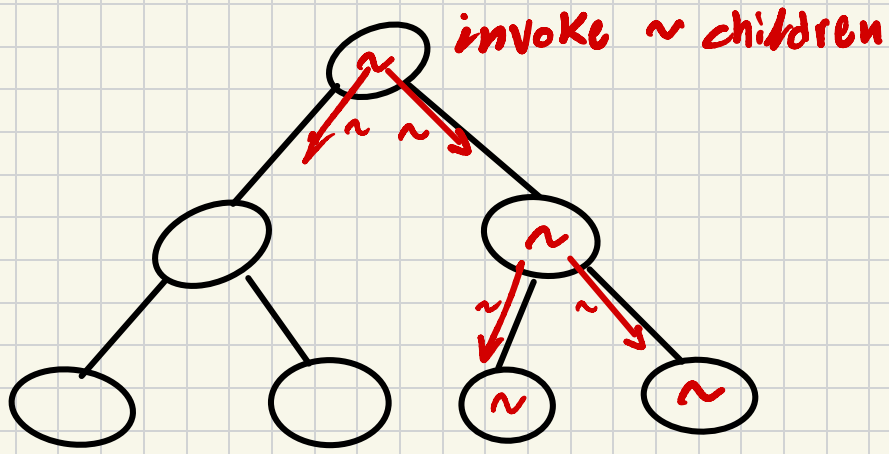
herdId
size

shared_ptr HorseTree

Full/empty Herds
m m0



$\sim \text{Tree}$
 $\Theta(n)$



all horses

reset
run

* static int for search call
* so if it don't match reset + Sync

* go over each node + 1 counter master

* Find node with no slaves

* Find node with no master

* start going forward and counting

- till nullptr

- till counted too much

- till got to master

if count $\neq$ N true

else False

